

THE HONG KONG UNIVERSITY OF SCIENCE & TECHNOLOGY
ARIN5103: Foundations of Machine Learning
Fall 2025 Assignment 2

Yang, Fengshuo/21216054

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Q1: SVM

(a)

Support vectors are defined as the points with non-zero Lagrange multipliers, means $\alpha_i > 0$.

Given:

$$\alpha_1 = 0.25, \quad \alpha_2 = 0, \quad \alpha_3 = 0.25$$

So the support vectors are:

$$\mathbf{x}_1 = (1, 1)^T \quad \text{and} \quad \mathbf{x}_3 = (0, 0)^T$$

(b)

- For $\mathbf{x}_1$ (positive support vector, $y_1 = +1$): $\sum_{j=1}^3 \alpha_j y_j k(\mathbf{x}_j, \mathbf{x}_1) + b = 1$
- For $\mathbf{x}_3$ (negative support vector, $y_3 = -1$): $\sum_{j=1}^3 \alpha_j y_j k(\mathbf{x}_j, \mathbf{x}_3) + b = -1$ where the kernel function is $k(\mathbf{x}_j, \mathbf{x}_i) = (\mathbf{x}_j^T \mathbf{x}_i + 1)^2$.

For $\mathbf{x}_1 = (1, 1)^T$:

$$- k(\mathbf{x}_1, \mathbf{x}_1) = ((1, 1)(1, 1)^T + 1)^2 = (3)^2 = 9$$

$$- k(\mathbf{x}_2, \mathbf{x}_1) = ((2, 2)(1, 1)^T + 1)^2 = (5)^2 = 25$$

$$- k(\mathbf{x}_3, \mathbf{x}_1) = ((0, 0)(1, 1)^T + 1)^2 = (1)^2 = 1$$

$$\text{Substitute: } 0.25 \cdot 1 \cdot 9 + 0 \cdot 1 \cdot 25 + 0.25 \cdot (-1) \cdot 1 + b = 2.25 - 0.25 + b = 1 \implies 2 + b = 1 \implies b = -1$$

Result:

$$b = -1$$

(c)

The decision function is:

$$f(\mathbf{x}) = \sum_{i=1}^3 \alpha_i y_i k(\mathbf{x}_i, \mathbf{x}) + b$$

For $\mathbf{x} = (x_a, x_b)^T$ and kernel $k(\mathbf{x}_i, \mathbf{x}) = (\mathbf{x}_i^T \mathbf{x} + 1)^2$:

- Term for $\mathbf{x}_1$: $\alpha_1 y_1 k(\mathbf{x}_1, \mathbf{x}) = 0.25 \cdot 1 \cdot (x_a + x_b + 1)^2$
- Term for $\mathbf{x}_2$: $\alpha_2 y_2 k(\mathbf{x}_2, \mathbf{x}) = 0$
- Term for $\mathbf{x}_3$: $\alpha_3 y_3 k(\mathbf{x}_3, \mathbf{x}) = 0.25 \cdot (-1) \cdot 1 = -0.25$

And bias $b = -1$:

$$\begin{aligned}
 f(\mathbf{x}) &= 0.25(x_a + x_b + 1)^2 - 0.25 - 1 \\
 &= 0.25(x_a^2 + 2x_a x_b + 2x_a + x_b^2 + 2x_b + 1) - 1.25 \\
 &= 0.25x_a^2 + 0.5x_a x_b + 0.5x_a + 0.25x_b^2 + 0.5x_b - 1
 \end{aligned}$$

Result:

$$0.25x_a^2 + 0.5x_a x_b + 0.5x_a + 0.25x_b^2 + 0.5x_b - 1$$

(d)

For $\mathbf{x}_{\text{new}} = (1, 0)^T$, substitute $x_a = 1$, $x_b = 0$ into the result from (c):

$$\begin{aligned}
 f(\mathbf{x}_{\text{new}}) &= 0.25(1)^2 + 0.5(1)(0) + 0.5(1) + 0.25(0)^2 + 0.5(0) - 1 \\
 &= 0.25 + 0 + 0.5 + 0 + 0 - 1 = -0.25
 \end{aligned}$$

determined by the sign:

$$\text{sign}(-0.25) = -1$$

Result:

$$-1$$